MoS2-SiO2 core-shell TENG microstructures

"what are the effect of using MoS2 as shell and SiO2 as core in TENG? what happened to their porosity (amount, size, distributions)? what happened on the morphology ? and the other important structural and microstructural factors ?" *Generated by LeapSpace*

MoS_2 as a **shell** on **SiO_2 cores** in TENGs is reported to give a **uniform, thin, hydrophobic outer layer** that improves **charge stability, humidity resistance, and durability**, but the retrieved evidence does **not** provide direct quantitative porosity data for the MoS_2@SiO_2 TENG case itself [11].

What happens in the MoS_2-shell / SiO_2-core TENG architecture

Aspect	Observed / supported effect	What it means for TENGs
Morphology	A **thin MoS_2 layer grows uniformly on SiO_2 nanoparticles** [11]	The SiO_2 core is effectively "encapsulated" by MoS_2, creating a conformal core-shell particle rather than a patchy coating [11]
Surface character	The MoS_2 shell is described as **stable and hydrophobic** [11]	This helps block water uptake and keeps the triboelectric surface usable in humid environments [11]
Charge behavior	The shell **stabilizes charge density** and **significantly reduces charge decay** [11]	Output becomes more stable over time, especially under high humidity [11]
Device durability	The TENG retained **more than 70% output over 25 h at 99% RH** [11]	The architecture is strongly humidity-resistant compared with an unprotected surface [11]

Porosity: what can and cannot be concluded

- **Direct evidence for MoS_2@SiO_2 TENG porosity is not provided** in the retrieved material [11].
- **Inference:** a conformal MoS_2 shell would likely **cover part of the exposed SiO_2 surface and reduce accessible outer-surface porosity**, but that specific effect on pore **amount, size, and distribution** is **not measured** in the TENG source [11].
- In related core-shell systems, shell growth can either **preserve a porous interior** while altering the outer surface, or **shift the pore distribution** depending on shell thickness and continuity, so porosity changes are architecture-dependent rather than fixed [55] [56].

Structural factors that matter most

The evidence points to four key structural controls for this type of TENG composite:

1. **Shell continuity and uniformity** — the MoS_2 shell is reported as uniformly grown, which is important for consistent charge behavior [11].
2. **Shell thickness** — a **thin** shell is emphasized, suggesting that too thick a coating could change transport and surface contact behavior [11].
3. **Interfacial shielding** — the MoS_2 layer protects the SiO_2 interface from water penetration, which is central to humidity resistance [11].
4. **Surface chemistry / hydrophobicity** — the outer MoS_2 surface helps maintain charge density and slow charge decay under humid conditions [11].

Bottom line

For **MoS_2 shell on SiO_2 core** in TENGs, the main benefit is **not pore creation** but **surface encapsulation**: the shell makes the particles more **hydrophobic, charge-stable, and humidity-resistant**, which improves TENG durability and output retention [11]. The evidence does **not** directly quantify porosity changes for this exact architecture, so any pore-size or pore-distribution change should be treated as an inference rather than a measured result [11].

Confidence level: **Medium** — there is direct TENG evidence for the MoS_2-encapsulated SiO_2 architecture and its functional effects, but not direct porosity measurements for that exact structure.

References

11. Kim, D.-H., Park, J.Y., Choi, H.S., (...), Baik, J.M. (2025) Advancing Humidity-Resistant Triboelectric Nanogenerators Through MoS_2-Encapsulated SiO_2 Nanoparticles for Self-Powered Gas Sensing Applications. *Advanced Energy Materials*. https://doi.org/10.1002/aenm.202405278

55. Chen, Y., Sun, W., Zheng, H., (...), Hao, C. (2021) Electrorheological response behavior of H2Ti2O5@MoS2@SiO2 core-shell nanoparticles. *Ceramics International*. https://doi.org/10.1016/j.ceramint.2021.05.118

56. Soni, V., Patial, S., Kumar, A., (...), Nguyen, V.-H. (2023) Covalent organic frameworks (COFs) core@shell nanohybrids: Novel nanomaterial support towards environmental sustainability applications. *Environmental Research*. https://doi.org/10.1016/j.envres.2023.116353